CO3 – Lattices and Boolean Algebra

Detailed Lecture Notes

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1. Fundamentals of Set Theory

1.1. Topics Covered

- Definition of sets
- Representation of sets
- Membership relation
- Cardinality
- Universal and null sets
- Subsets and proper subsets
- Set operations
- Cartesian product
- Power set
- Partition of sets
- De Morgan's laws

Learning Objectives

- Understand the basic language and notation of set theory.
- Learn how to represent and manipulate sets mathematically.
- Develop intuition regarding subsets, unions, intersections and complements.
- Apply set operations in elementary mathematical reasoning.
- Prepare the foundation for relations, functions and higher mathematics.

1.2. Definition of Sets

Definition 1.1. *A **set** is a well-defined collection of distinct objects.*

Example 1.2.

$$A = \{1, 2, 3, 4\}$$

$$B = \{a, e, i, o, u\}$$

1.3. Representation of Sets

Sets may be represented in two forms:

1. Roster form
2. Set-builder form

Example 1.3. *Roster form:*

$$A = \{2, 4, 6, 8\}$$

Set-builder form:

$$A = \{x : x \text{ is an even positive integer less than } 10\}$$

1.4. Membership Relation

If an element a belongs to a set A, we write

$$a \in A$$

If a does not belong to A, we write

$$a \notin A$$

1.5. Cardinality of a Set

Definition 1.4. *The number of elements in a finite set is called its* ***cardinality****.*

If A is a finite set, then the cardinality of A is denoted by

$$|A|$$

1.6. Universal Set and Null Set

Definition 1.5. *The* ***universal set*** *is the set containing all objects under consideration.*

Definition 1.6. *A set containing no elements is called the* ***null set*** *or empty set. It is denoted by*

$$\emptyset$$

1.7. Subsets and Proper Subsets

Definition 1.7. *A set A is said to be a subset of B if every element of A belongs to B.*

$$A \subseteq B$$

Definition 1.8. *A set A is called a proper subset of B if*

$$A \subseteq B$$

and

$$A \neq B$$

This is denoted by

$$A \subset B$$

1.8. Set Operations

Union

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

Intersection

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

Complement

$$A' = \{x : x \in U \text{ and } x \notin A\}$$

1.9. Cartesian Product

Definition 1.9. *The Cartesian product of two sets A and B is defined as*

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

1.10. Power Set

Definition 1.10. *The set of all subsets of a set A is called the* ***power set*** *of A.*

1.11. Partition of a Set

Definition 1.11. *A partition of a set A is a collection of non-empty disjoint subsets whose union is A.*

1.12. De Morgan's Laws for Sets

Theorem 1.12 (De Morgan's Laws)**.**

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

Partial Order Relations and Posets

- Introduction to partial order relations
- Definition of partial order relations
- Mathematical conditions for partial orders
- Partially ordered sets (posets)
- Examples of partial order relations
- Non-partial order relations
- Comparable and incomparable elements
- Hasse diagrams
- Construction rules for Hasse diagrams
- Minimal and maximal elements
- Least and greatest elements
- Upper bounds and lower bounds
- Least upper bound and greatest lower bound
- Matrix representation of partial orders
- Applications of partial order relations

Learning Objectives

- Understand the concept of partial order relations.
- Determine whether a given relation is a partial order relation.
- Understand the structure of partially ordered sets (posets).
- Distinguish between comparable and incomparable elements.
- Differentiate partial orders from total orders.
- Construct and interpret Hasse diagrams.
- Identify minimal, maximal, least and greatest elements in posets.
- Determine upper bounds and lower bounds of subsets.
- Compute least upper bounds and greatest lower bounds.
- Analyze partial order relations using matrices and graphical representations.

2. Partial Order Relations and Posets

2.1. Introduction

Many real-world systems involve ordering or arrangement of objects.

Examples include:

- prerequisite structures,
- divisibility,
- subset inclusion,
- organizational hierarchy,
- dependency networks.

Such structures are modeled using partial order relations.

2.2. Partial Order Relation

Definition 2.1. *A relation R on a set A is called a* ***partial order relation*** *if it is:*

- *reflexive,*
- *antisymmetric,*
- *transitive.*

2.3. Mathematical Conditions

A relation R on A is a partial order relation if:

$$(a, a) \in R$$

for every:

$$a \in A$$

and

$$(a, b) \in R$$

and

$$(b, a) \in R$$

imply:

$$a = b$$

and

$$(a, b) \in R$$

and

$$(b, c) \in R$$

imply:

$$(a, c) \in R$$

2.4. Partially Ordered Set (Poset)

Definition 2.2. *A set together with a partial order relation is called a partially ordered set or* ***poset****.*

It is denoted by:

$$(A, R)$$

2.5. Examples of Partial Orders

Example 2.3. *Subset relation:*

$$\subseteq$$

on the power set:

$$\mathcal{P}(A)$$

is a partial order relation.

Example 2.4. *Divisibility relation:*

$$|$$

on natural numbers is a partial order relation.

Example 2.5. *The relation:*

$$\leq$$

on real numbers is a partial order relation.

Example 2.6. *Let:*

$$A = \{1, 2, 3, 6\}$$

Define relation:

$$aRb \iff a|b$$

Then:

$$R = \{(1,1), (1,2), (1,3), (1,6), (2,2), (2,6), (3,3), (3,6), (6,6)\}$$

This relation is:

- *reflexive,*
- *antisymmetric,*
- *transitive.*

Hence it is a partial order relation.

Worked-Out Examples

Example 2.7. *Check if the relation defined by $\preceq$ defined on the set A is a partial order relation.*
On $A = \mathbb{N}$, define $a \preceq b$ if $a \leq b$.

Proof:

Reflexive: $a \leq a$.
Antisymmetric: If $a \leq b$ and $b \leq a$, then $a = b$.
Transitive: If $a \leq b$ and $b \leq c$, then $a \leq c$.
Hence, $\preceq$ is a partial order.

Example 2.8. *Check if the relation defined by $\preceq$ defined on the set A is a partial order relation.*
On $A = \mathcal{P}(\{1, 2, 3\})$, define $X \preceq Y$ if $X \subseteq Y$.

Proof:

Reflexive: $X \subseteq X$.
Antisymmetric: If $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$.
Transitive: If $X \subseteq Y$ and $Y \subseteq Z$, then $X \subseteq Z$.
Hence, $\preceq$ is a partial order.

Example 2.9. *Check if the relation defined by $\preceq$ defined on the set A is a partial order relation.*
On $A = \mathbb{Z}$, define $a \preceq b$ if a divides b.

Proof:

Reflexive: a divides a.
Antisymmetric: If $a|b$ and $b|a$, then $a = \pm b$. Restricting to $\mathbb{N}$ gives $a = b$.
Transitive: If $a|b$ and $b|c$, then $a|c$.
Hence, $\preceq$ is a partial order.

Example 2.10. *Check if the relation defined by $\preceq$ defined on the set A is a partial order relation.*
On $A = \mathbb{R}^n$, define $u \preceq v$ if $u_i \leq v_i$ for all coordinates i.

Proof:

Reflexive: $u_i \leq u_i$.

Antisymmetric: If $u_i \leq v_i$ *and* $v_i \leq u_i$ *for all* i, *then* $u = v$.

Transitive: If $u \preceq v$ *and* $v \preceq w$, *then* $u \preceq w$.

Hence, $\preceq$ *is a partial order.*

Example 2.11. *Check if the relation defined by* $\preceq$ *defined on the set* A *is a partial order relation.*

On $A =$ *set of tasks with precedence constraints, define* $T_1 \preceq T_2$ *if* T_1 *must be completed before* T_2.

Proof:

Reflexive: Each task precedes itself trivially.

Antisymmetric: If T_1 *precedes* T_2 *and* T_2 *precedes* T_1, *then* $T_1 = T_2$.

Transitive: If T_1 *precedes* T_2 *and* T_2 *precedes* T_3, *then* T_1 *precedes* T_3.

Hence, $\preceq$ *is a partial order.*

Example 2.12. *Check if the relation defined by* $\preceq$ *defined on the set* A *is a partial order relation.*

On $A =$ *set of positive integers, define* $a \preceq b$ *if* $\gcd(a, b) = a$.

Proof:

Reflexive: $\gcd(a, a) = a$.

Antisymmetric: If $\gcd(a, b) = a$ *and* $\gcd(b, a) = b$, *then* $a = b$.

Transitive: If $\gcd(a, b) = a$ *and* $\gcd(b, c) = b$, *then* $\gcd(a, c) = a$.

Hence, $\preceq$ *is a partial order.*

Example 2.13. *Check if the relation defined by* $\preceq$ *defined on the set* A *is a partial order relation.*

On $A = \mathbb{R}$, *define* $a \preceq b$ *if* $|a| \leq |b|$.

Proof:

Reflexive: $|a| \leq |a|$.

Antisymmetric: If $|a| \leq |b|$ *and* $|b| \leq |a|$, *then* $|a| = |b|$. *But* a *and* b *may differ by sign, so this is not antisymmetric unless restricted to* $\mathbb{R}_{\geq 0}$.

On $\mathbb{R}_{\geq 0}$, $\preceq$ *is a partial order.*

Example 2.14. *Check if the relation defined by* $\preceq$ *defined on the set* A *is a partial order relation.*

On $A =$ *set of strings over* $\{0, 1\}$, *define* $u \preceq v$ *if* u *is a prefix of* v.

Proof:

Reflexive: Every string is a prefix of itself.

Antisymmetric: If u is a prefix of v and v is a prefix of u, then $u = v$.
Transitive: If u prefix of v and v prefix of w, then u prefix of w.
Hence, $\preceq$ is a partial order.

Example 2.15. *Check if the relation defined by $\preceq$ defined on the set A is a partial order relation.*
On A = set of subspaces of a vector space V, define $U \preceq W$ if $U \subseteq W$.
Proof:
Reflexive: $U \subseteq U$.
Antisymmetric: If $U \subseteq W$ and $W \subseteq U$, then $U = W$.
Transitive: If $U \subseteq W$ and $W \subseteq Z$, then $U \subseteq Z$.
Hence, $\preceq$ is a partial order.

Example 2.16. *Check if the relation defined by $\preceq$ defined on the set A is a partial order relation.*
On A = set of equivalence relations on a set S, define $R_1 \preceq R_2$ if $R_1 \subseteq R_2$.
Proof:
Reflexive: $R_1 \subseteq R_1$.
Antisymmetric: If $R_1 \subseteq R_2$ and $R_2 \subseteq R_1$, then $R_1 = R_2$.
Transitive: If $R_1 \subseteq R_2$ and $R_2 \subseteq R_3$, then $R_1 \subseteq R_3$.
Hence, $\preceq$ is a partial order.

2.6. Non-Partial Order Relation

Example 2.17. *Friendship relation is not a partial order relation because it is symmetric and violates antisymmetry.*

2.7. Comparable Elements

Definition 2.18. *Two elements:*

$$a, b \in A$$

are said to be comparable if:

$$aRb$$

or

$$bRa$$

2.8. Incomparable Elements

Definition 2.19. *Two elements are incomparable if neither:*

$$aRb$$

nor:

$$bRa$$

holds.

Example 2.20. *Under subset relation:*

$$\{1\}$$

and

$$\{2\}$$

are incomparable because:

$$\{1\} \not\subseteq \{2\}$$

and

$$\{2\} \not\subseteq \{1\}$$

Example 2.21. *The set of integers with relation:*

$$\leq$$

forms a totally ordered set.

2.9. Hasse Diagram

Definition 2.22. *A Hasse diagram is a graphical representation of a finite partially ordered set.*

2.10. Construction Rules for Hasse Diagram

To construct a Hasse diagram:

- remove self-loops,
- remove transitive edges,
- place smaller elements below larger elements.

2.11. Example of Hasse Diagram

Example 2.23. *Draw the Hasse diagram of the poset* $(\mathcal{P}(X), \subseteq)$*, where* $X = \{a, b, c\}$.

2.12. Minimal and Maximal Elements

Definition 2.24. *An element* $a \in A$ *is minimal if no distinct element is smaller than* a.

Definition 2.25. *An element* $b \in A$ *is maximal if no distinct element is greater than* b.

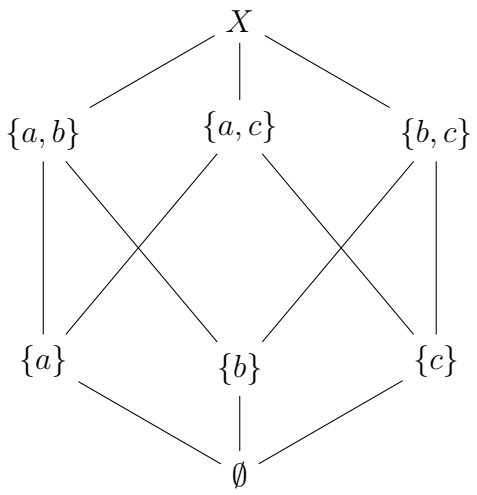


Figure 1: Iteration 1: Hasse diagram

Example 2.26. *In the divisibility poset:*

$$A = \{1, 2, 3, 6\}$$

- *minimal element is:*

$$1$$

- *maximal element is:*

$$6$$

2.13. Least and Greatest Elements

Definition 2.27. *An element $a \in A$ is the least element if:*

$$aRb$$

for every:

$$b \in A$$

Definition 2.28. *An element $g \in A$ is the greatest element if:*

$$bRg$$

for every:

$$b \in A$$

Example 2.29. *In:*

$$A = \{1, 2, 3, 6\}$$

under divisibility:

- *least element is:*

$$1$$

- *greatest element is:*

$$6$$

2.14. Upper Bound and Lower Bound

Definition 2.30. *An element u is an upper bound of a subset:*

$$B \subseteq A$$

if:

$$bRu$$

for all:

$$b \in B$$

Definition 2.31. *An element l is a lower bound of a subset:*

$$B \subseteq A$$

if:

$$lRb$$

for all:

$$b \in B$$

2.15. Least Upper Bound and Greatest Lower Bound

Definition 2.32. *The least upper bound is called the supremum or join.*

Definition 2.33. *The greatest lower bound is called the infimum or meet.*

Example 2.34. *For subset relation:*

$$\{1\}, \{2\}$$

have:

- *least upper bound:*

$$\{1, 2\}$$

- *greatest lower bound:*

$$\emptyset$$

2.16. Matrix Representation of Partial Orders

Note 2.35. *A matrix representation of a partial order relation must satisfy:*

- *reflexivity,*
- *antisymmetry,*
- *transitivity.*

Example 2.36.

$$M_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.17. Important Observations

- Every total order is a partial order.
- Every partial order need not be total.
- Posets may contain incomparable elements.
- Hasse diagrams simplify visualization of finite posets.

2.18. Applications of Partial Orders

- dependency scheduling,
- compiler design,
- hierarchy systems,
- database theory,
- lattice theory,
- graph theory.

2.19. Conclusion

Partial order relations and posets form the foundation for:

- order theory,
- lattice theory,
- algebraic structures,

- graph theory,
- optimization,
- and advanced discrete mathematics.

Hasse Diagrams & Lattices

- Review of partial order relations
- Hasse diagrams
- Rules for constructing Hasse diagrams
- Minimal and maximal elements
- Least and greatest elements
- Upper bounds and lower bounds
- Least upper bound and greatest lower bound
- Lattice structures
- Join and meet operations
- Bounded lattices
- Complemented lattices
- Distributive lattices
- Boolean algebra
- Applications of Hasse diagrams and lattices

Learning Objectives

- Understand graphical representation of partially ordered sets using Hasse diagrams.
- Learn the rules for constructing Hasse diagrams.
- Identify minimal, maximal, least and greatest elements in posets.
- Determine upper bounds and lower bounds of subsets.
- Compute least upper bounds and greatest lower bounds.
- Understand the concept of lattices and their mathematical properties.
- Study join and meet operations in lattices.
- Distinguish bounded, complemented and distributive lattices.
- Understand the structure of Boolean algebras.
- Analyze lattice structures arising from power sets and divisibility relations.

3. Hasse Diagrams, Lattices and Advanced Posets

3.1. Introduction

Partially ordered sets can often be represented graphically in a simplified form.

This graphical representation:

- improves visualization,
- simplifies order relationships,
- helps identify bounds and lattice structures.

3.2. Review of Partial Orders

Definition 3.1. *A relation R on a set A is a partial order relation if it is:*

- *reflexive,*
- *antisymmetric,*
- *transitive.*

3.3. Hasse Diagram

Definition 3.2. *A Hasse diagram is a simplified graphical representation of a finite partially ordered set.*

3.4. Rules for Constructing Hasse Diagrams

To construct a Hasse diagram:

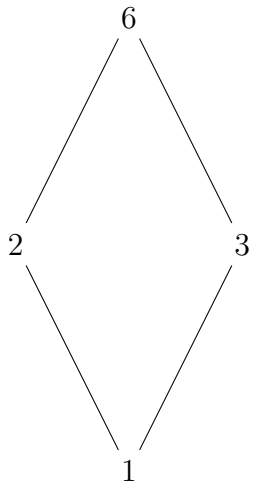
1. Remove all self-loops.
2. Remove all transitive edges.
3. Place smaller elements below larger elements.
4. Connect comparable elements using upward edges.

3.5. Example of Hasse Diagram

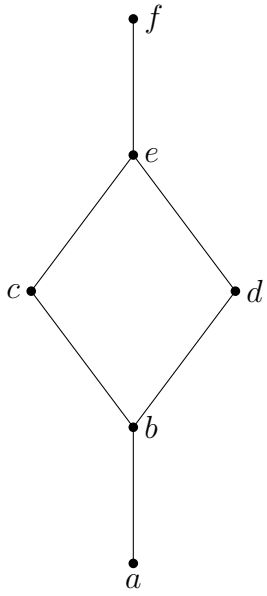
Example 3.3. *Consider:*

$$A = \{1, 2, 3, 6\}$$

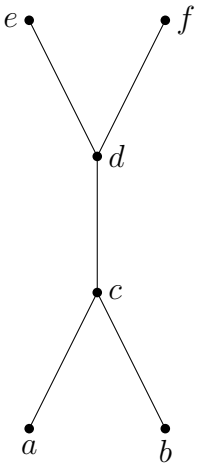
under divisibility relation.



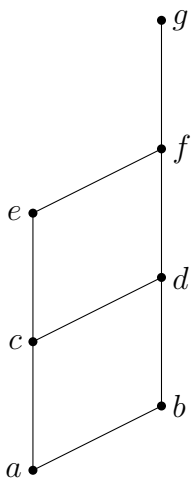
Example 3.4. *Determine whether the posets represented by the Hasse diagrams given in Figure below are lattices.*



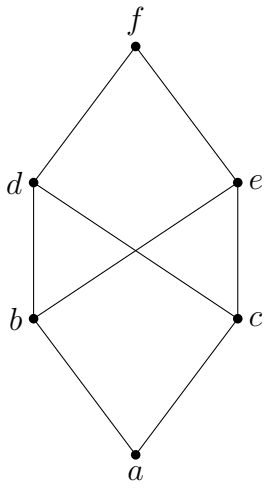
(a)



(b)



(c)



(d)

(a) The poset represented by the Hasse diagram in Fig.(a) is a lattice, since every pair of elements of this poset has both an LUB and a GLB.

(b) The pair of elements a, b does not have a GLB and the pair e, f does not have an LUB. Hence the poset in Fig.(b) is not a lattice.

(c) Since every pair of elements of the poset in Fig.(c) has both an LUB and a GLB, it is a lattice.

(d) Though the pair of elements $\{b, c\}$ has three upper bounds d, e, f, none of these precedes the other two, i.e. $\{b, c\}$ does not have an LUB. Hence the poset in Fig.(d) is not a lattice.

3.6. Minimal and Maximal Elements

Definition 3.5. *An element $a \in A$ is minimal if no distinct element precedes it.*

Definition 3.6. *An element $b \in A$ is maximal if no distinct element succeeds it.*

Example 3.7. *In:*

$$A = \{1, 2, 3, 6\}$$

under divisibility:

- *minimal element:*

$$1$$

- *maximal element:*

$$6$$

3.7. Least and Greatest Elements

Definition 3.8. *An element $l \in A$ is the least element if:*

$$lRa$$

for every:

$$a \in A$$

Definition 3.9. *An element $g \in A$ is the greatest element if:*

$$aRg$$

for every:

$$a \in A$$

Note 3.10. *A least element and a greatest element are unique whenever they exist.*

3.8. Upper Bounds and Lower Bounds

Definition 3.11. *An element u is an upper bound of:*

$$B \subseteq A$$

if:

$$bRu$$

for all:

$$b \in B$$

Definition 3.12. *An element l is a lower bound of:*

$$B \subseteq A$$

if:

$$lRb$$

for all:

$$b \in B$$

3.9. Least Upper Bound and Greatest Lower Bound

Definition 3.13. *The least upper bound of a subset is called:*

- *supremum,*
- *join.*

Definition 3.14. *The greatest lower bound of a subset is called:*

- *infimum,*
- *meet.*

Example 3.15. *Under subset relation:*

$$\{1\}, \{2\}$$

have:

- *least upper bound:*

$$\{1, 2\}$$

- *greatest lower bound:*

$$\emptyset$$

3.10. Lattice

Definition 3.16. *A partially ordered set is called a lattice if every pair of elements has:*

- *a least upper bound,*
- *a greatest lower bound.*

3.11. Examples of Lattices

Example 3.17. *If $P(S)$ is the power set of a set S and $\cup$ and $\cap$ are taken as the join and meet, prove that $\{P(S), \subseteq\}$ is a lattice.*

Let A and B be any two elements of $P(S)$, i.e. any two subsets of S.
Then an upper bound of $\{A, B\}$ is a subset of S that contains both A and B and the least among them is $A \cup B \in P(S)$, as can be seen from the following:

$$A \subseteq A \cup B \quad \text{and} \quad B \subseteq A \cup B$$

i.e. $A \cup B$ is an upper bound of $\{A, B\}$.
If we assume that

$$A \subseteq C \quad \text{and} \quad B \subseteq C,$$

then

$$A \cup B \subseteq C.$$

Thus the LUB of $\{A, B\}$ is

$$A \vee B = A \cup B.$$

Similarly,

$$A \cap B \subseteq A \quad \text{and} \quad A \cap B \subseteq B$$

i.e. $A \cap B$ is a lower bound of $\{A, B\}$.
If we assume that

$$C \subseteq A \quad \text{and} \quad C \subseteq B,$$

then

$$C \subseteq A \cap B.$$

Thus the GLB of $\{A, B\}$ is

$$A \wedge B = A \cap B.$$

Hence every pair of elements of $P(S)$ has both an LUB and a GLB under set inclusion relation.

Therefore,

$$\{P(S), \subseteq\}$$

is a lattice.

Example 3.18. *The set of divisors of a number under divisibility relation forms a lattice.*

3.12. Join and Meet Operations

Definition 3.19. *For elements:*

$$a, b \in A$$

their:

- *join is denoted by:*

$$a \vee b$$

- *meet is denoted by:*

$$a \wedge b$$

Example 3.20. *In:*

$$\mathcal{P}(\{1, 2\})$$

under subset relation:

$$\{1\} \vee \{2\} = \{1, 2\}$$

and:

$$\{1\} \wedge \{2\} = \emptyset$$

3.13. Bounded Lattice

Definition 3.21. *A lattice having:*

- *a least element,*
- *a greatest element*

is called a bounded lattice.

3.14. Complemented Lattice

Definition 3.22. *A bounded lattice is complemented if every element has a complement.*

Example 3.23. *Prove that*

$$D_{42} = \{S_{42}, D\}$$

is a complemented lattice by finding the complements of all the elements.

$$D_{42} = \{1, 2, 3, 6, 7, 14, 21, 42\}$$

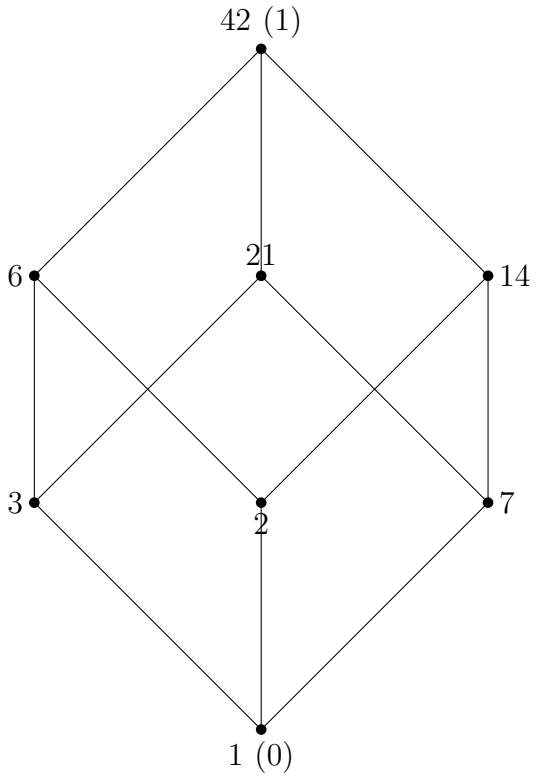
The Hasse diagram of D_{42} is given in Figure below.
The zero element of the lattice is 1 *and the unit element of the lattice is* 42.

$$1 \vee 42 = \text{LCM}\{1, 42\} = 42 \equiv 1'$$

and

$$1 \wedge 42 = \text{GCD}\{1, 42\} = 1 \equiv 0$$

$$\therefore \qquad 1' = 42$$



Similarly we can find that

$$2' = 21, \qquad 3' = 14, \qquad 6' = 7,$$

$$7' = 6, \qquad 14' = 3, \qquad 21' = 2$$

and

$$42' = 1.$$

Since every element of D_{42} has a complement, it is a complemented lattice.

3.15. Distributive Lattice

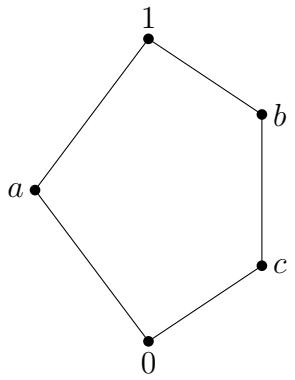
Definition 3.24. *A lattice is distributive if:*

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

and:

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

Example 3.25. *Verify whether the lattice given by the Hasse diagram in Figure is distributive.*



$$a \wedge (b \vee c) = a \wedge b = 0$$

Also,

$$(a \wedge b) \vee (a \wedge c) = 0 \vee 0 = 0$$

$$\therefore \quad a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c) \tag{1}$$

Now,

$$c \wedge (a \vee b) = c \wedge 1 = c$$

Also,

$$(c \wedge a) \vee (c \wedge b) = 0 \vee c = c$$

$$\therefore \quad c \wedge (a \vee b) = (c \wedge a) \vee (c \wedge b) \tag{2}$$

Steps (1) and (2) do not mean that the lattice is distributive.

Now let us consider

$$b \wedge (c \vee a) = b \wedge 1 = b$$

But

$$(b \wedge c) \vee (b \wedge a) = c \vee 0 = c$$

Therefore,

$$b \wedge (c \vee a) \neq (b \wedge c) \vee (b \wedge a)$$

Hence the given lattice is not distributive.

3.16. Boolean Algebra

Definition 3.26. *A complemented distributive lattice is called a Boolean algebra.*

3.17. Example of Boolean Algebra

Example 3.27. *If $P(S)$ is the power set of a non-empty set S, prove that*

$$\{P(S), \cup, \cap, \setminus, \phi, S\}$$

is a Boolean algebra, where the complement of any set

$$A \subseteq S$$

is taken as

$$S \setminus A$$

or

$$S - A$$

that is the relative complement of A with respect to S.

Let X, Y and Z be any three elements of $P(S)$.

$$X \cup \phi = X \qquad \text{and} \qquad X \cap S = X$$

Thus ϕ and S play the roles of 0 and 1 and the identity laws are satisfied. *(1)*

$$X \cup Y = Y \cup X \qquad \text{and} \qquad X \cap Y = Y \cap X$$

i.e. the commutative laws are satisfied. *(2)*

$$(X \cup Y) \cup Z = X \cup (Y \cup Z)$$

and

$$(X \cap Y) \cap Z = X \cap (Y \cap Z)$$

i.e. the associative laws hold good. *(3)*

$$X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$$

and

$$X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$$

i.e. the distributive laws hold good. *(4)*

$$X \cup (S - X) = S$$

and

$$X \cap (S - X) = \phi$$

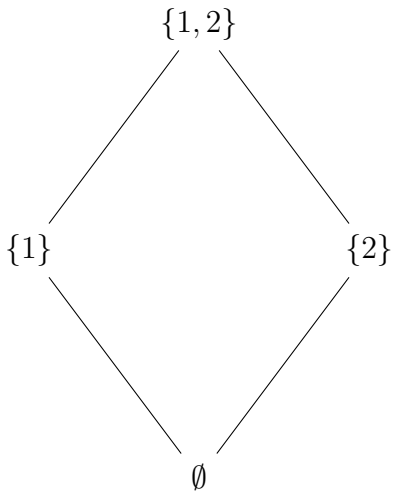
i.e. the complement laws hold good. *(5)*

Thus all the five axioms of Boolean algebra hold good.
Hence

$$\{P(S), \cup, \cap, \setminus, \phi, S\}$$

is a Boolean algebra.

3.18. Visual Example of a Lattice



3.19. Applications of Hasse Diagrams and Lattices

- compiler optimization,
- data organization,
- logic design,
- switching circuits,
- digital electronics,
- algebraic structures,
- knowledge representation.

3.20. Important Observations

- Every lattice is a poset.
- Every poset need not be a lattice.
- Hasse diagrams simplify finite posets.
- Boolean algebras arise naturally from power sets.

3.21. Exercises

1. Construct Hasse diagrams for finite posets.
2. Identify:
    - minimal elements,

- maximal elements,
- least element,
- greatest element.

3. Find:
   - upper bounds,
   - lower bounds,
   - supremum,
   - infimum.
4. Determine whether the given poset is a lattice.
5. Find join and meet of elements.
6. Determine whether a lattice is:
   - bounded,
   - complemented,
   - distributive.
7. Verify whether a given structure forms a Boolean algebra.

3.22. Conclusion

Hasse diagrams and lattice structures form the foundation for:

- order theory,
- Boolean algebra,
- switching theory,
- digital logic,
- abstract algebra,
- and advanced discrete mathematics.

3.23. Lattice Laws

Theorem 3.28 (Commutative Laws)**.**

$$a \vee b = b \vee a$$

$$a \wedge b = b \wedge a$$

Theorem 3.29 (Associative Laws)**.**

$$a \vee (b \vee c) = (a \vee b) \vee c$$

$$a \wedge (b \wedge c) = (a \wedge b) \wedge c$$

Theorem 3.30 (Absorption Laws)**.**

$$a \vee (a \wedge b) = a$$

$$a \wedge (a \vee b) = a$$

4. Complemented Lattice

Definition 4.1. *A bounded lattice is called a* ***complemented lattice*** *if every element has a complement.*

If 0 and 1 are the least and greatest elements respectively, then b is called a complement of a if

$$a \vee b = 1$$

and

$$a \wedge b = 0$$

Example 4.2. *The power set lattice* $(P(S), \cup, \cap)$ *is complemented.*

5. Distributive Lattice

Definition 5.1. *A lattice is called distributive if*

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

and

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

Example 5.2. *The lattice* $(P(S), \cup, \cap)$ *is distributive.*

Example 5.3. *Every Boolean algebra is a distributive lattice.*

6. Boolean Algebra

Definition 6.1. *A Boolean algebra is an algebraic structure*

$$(B, +, \cdot, ', 0, 1)$$

satisfying:

1. *Closure laws*
2. *Commutative laws*
3. *Associative laws*
4. *Distributive laws*
5. *Identity laws*
6. *Complement laws*

6.1. Basic Laws of Boolean Algebra

Theorem 6.2 (Identity Laws)**.**

$$a + 0 = a$$

$$a \cdot 1 = a$$

Theorem 6.3 (Null Laws)**.**

$$a + 1 = 1$$

$$a \cdot 0 = 0$$

Theorem 6.4 (Idempotent Laws)**.**

$$a + a = a$$

$$a \cdot a = a$$

Theorem 6.5 (Complement Laws)**.**

$$a + a' = 1$$

$$a \cdot a' = 0$$

Theorem 6.6 (De Morgan's Laws)**.**

$$(a + b)' = a'b'$$

$$(ab)' = a' + b'$$

6.2. Principle of Duality

Definition 6.7. *The principle of duality states that every algebraic expression remains valid if:*

1. $+$ *and* $\cdot$ *are interchanged*
2. 0 *and* 1 *are interchanged*

7. Boolean Functions

Definition 7.1. *A Boolean function is a function*

$$f : \{0,1\}^n \to \{0,1\}$$

Example 7.2.

$$f(x,y) = x + y$$

$$f(x,y) = xy$$

$$f(x,y,z) = x'y + yz$$

7.1. Truth Table

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

8. Normal Forms in Propositional Logic

What Are Normal Forms?

Turning any complex logical statement into a standard "recipe" so everyone – or every algorithm – follows the same steps every time.

Why Normal Forms Matter

- Provide a uniform structure for simplification and mechanical reasoning.
- Underpin algorithms like SAT-solvers and automated proof engines.
- Help us to see patterns and apply general rules systematically.

8.1. Conjunctive Normal Form (CNF)

When we are interested in listing all the ways a formula can avoid being false.

Definition 8.1. *Conjunctive Normal Form (CNF) is an "AND of OR-clauses."*

- Think of a checklist:

(item A OR item B) AND (item C OR item D OR item E)

- Example:

$$(p \vee \neg q) \wedge (r \vee s \vee \neg t)$$

8.2. Disjunctive Normal Form (DNF)

When we are interested in transforming a proposition into a logically equivalent expression that explicitly lists all the ways the formula can be true.

Definition 8.2. *Disjunctive Normal Form (DNF) is an "OR of AND-clauses."*

- Think of menu combos:

(dish 1 AND side A) OR (dish 2 AND side B)

- Example:

$$(p \wedge q) \vee (\neg p \wedge r \wedge s)$$

8.3. Truth Table Mapping

CNF: Built from rows where formula is false. Each row gives a clause that must be false to falsify the formula.

DNF: Built from rows where formula is true. Each row gives a term that must be true to satisfy the formula.

Example: Formula

$$p \to q$$

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

CNF: From row 2

$$(\neg p \vee q)$$

DNF: From rows 1, 3, 4

$$(p \wedge q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q)$$

9. Procedure to Obtain the DNF or CNF of a Given Formula

Step 1

If the connectives

$$\rightarrow \text{ and } \leftrightarrow$$

are present in the given formula they are replaced by

$$\wedge,\ \vee \text{ and } \neg$$

viz.

$$p \rightarrow q$$

is replaced by

$$\neg p \vee q$$

and

$$p \leftrightarrow q$$

is replaced by either

$$(p \wedge q) \vee (\neg p \wedge \neg q)$$

or

$$(\neg p \vee q) \wedge (\neg q \vee p).$$

Step 2

If the negation is present before the given formula or a part of the given formula (not a variable), De Morgan's laws are applied so that the negation is brought before the variables only.

Step 3

If necessary, the distributive law and the idempotent law are applied.

Step 4

If there is an elementary product which is equivalent to the truth value

$$F$$

in the DNF, it is omitted. Similarly if there is an elementary sum which is equivalent to the truth value

$$T$$

in the CNF, it is omitted.

For example, the DNF of

$$q \to (q \to p)$$

is given by

$$q \to (q \to p) \equiv \neg q \vee (q \to p)$$

$$\equiv \neg q \vee (\neg q \vee p)$$

$$\equiv (\neg q \vee \neg q) \vee p, \quad \text{by associative law}$$

$$\equiv \neg q \vee p, \quad \text{by idempotent law.}$$

The CNF of

$$\neg(p \vee q) \leftrightarrow (p \wedge q)$$

is given by

$$\neg(p \vee q) \leftrightarrow (p \wedge q)$$

$$\equiv (\neg(p \vee q) \wedge (p \wedge q)) \vee (\neg[\neg(p \vee q)] \wedge \neg(p \wedge q))$$

$$\equiv (\neg p \wedge \neg q) \wedge (p \wedge q) \vee (p \vee q) \wedge (\neg p \vee \neg q)$$

$$\equiv (p \wedge \neg p) \wedge (q \wedge \neg q) \vee (p \vee q) \wedge (\neg p \vee \neg q)$$

$$\equiv F \wedge F \vee (p \vee q) \wedge (\neg p \vee \neg q)$$

$$\equiv (p \vee q) \wedge (\neg p \vee \neg q)$$

Example 9.1. *Find the disjunctive normal forms of the following statements:*

1. $$\neg((p \leftrightarrow q) \wedge r)$$

2. $$p \vee (\neg p \rightarrow (q \vee (q \rightarrow \neg r)))$$

3. $$p \wedge \neg(q \wedge r) \vee (p \rightarrow q)$$

4. $$(p \wedge \neg(q \vee r)) \vee (((p \wedge q) \vee \neg r) \wedge p)$$

Solution (1)

$$\neg((p \leftrightarrow q) \wedge r)$$

$$\equiv \neg[((p \wedge q) \vee (\neg p \wedge \neg q)) \wedge r]$$

$$\equiv \neg[((p \wedge q) \wedge r) \vee ((\neg p \wedge \neg q) \wedge r)]$$

$$\equiv \neg[((p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r))]$$

$$\equiv \neg(p \wedge q \wedge r) \wedge \neg(\neg p \wedge \neg q \wedge r)$$

$$\equiv (\neg p \vee \neg q \vee \neg r) \wedge (p \vee q \vee \neg r)$$

Solution (2)

$$p \vee (\neg p \rightarrow (q \vee (q \rightarrow \neg r)))$$

$$\equiv p \vee (p \vee (q \vee (\neg q \vee \neg r)))$$

$$\equiv p \vee p \vee q \vee \neg q \vee \neg r$$

$$\equiv p \vee q \vee \neg q \vee \neg r$$

$$\equiv p \vee T \vee \neg r$$

$$\equiv T$$

Note 9.2. *The given statement is a tautology, as*

$$p \vee (q \vee \neg q) \vee \neg r \equiv p \vee T \vee \neg r \equiv T$$

Solution (3)

$$p \wedge \neg(q \wedge r) \vee (p \to q)$$

$$\equiv p \wedge (\neg q \vee \neg r) \vee (\neg p \vee q)$$

$$\equiv (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \vee q)$$

Solution (4)

$$(p \wedge \neg(q \vee r)) \vee (((p \wedge q) \vee \neg r) \wedge p)$$

$$\equiv (p \wedge \neg q \wedge \neg r) \vee ((p \wedge q \wedge p) \vee (\neg r \wedge p))$$

$$\equiv (p \wedge \neg q \wedge \neg r) \vee (p \wedge q) \vee (p \wedge \neg r)$$

Example 9.3. *Find the conjunction normal forms of the following statements:*

1. $$(p \wedge \neg(q \wedge r)) \vee (p \to q)$$
2. $$(q \vee (p \wedge q)) \wedge \neg((p \vee r) \wedge q)$$
3. $$(p \wedge \neg(q \vee r)) \vee (((p \wedge q) \vee \neg r) \wedge p)$$

Solution (1)

$$(p \wedge \neg(q \wedge r)) \vee (p \to q)$$

$$\equiv (p \wedge (\neg q \vee \neg r)) \vee (\neg p \vee q)$$

$$\equiv (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \vee q)$$

$$\equiv (p \vee \neg p \vee q) \wedge (\neg q \vee \neg p \vee q) \wedge (p \vee \neg r \vee \neg p \vee q)$$

$$\equiv T \wedge T \wedge (p \vee \neg r \vee q)$$

$$\equiv p \vee \neg r \vee q$$

Solution (2)

$$[q \vee (p \wedge q)] \wedge \neg[(p \vee r) \wedge q]$$

$$\equiv q \wedge \neg[(p \vee r) \wedge q], \quad \text{by absorption law}$$

$$\equiv q \wedge [\neg(p \vee r) \vee \neg q]$$

$$\equiv q \wedge [(\neg p \wedge \neg r) \vee \neg q]$$

$$\equiv q \wedge (\neg p \vee \neg q) \wedge (\neg q \vee \neg r)$$

Solution (3)

$$(p \wedge \neg(q \vee r)) \vee (((p \wedge q) \vee \neg r) \wedge p)$$

$$\equiv (p \wedge \neg q \wedge \neg r) \vee ((p \vee \neg r) \wedge (q \vee \neg r) \wedge p)$$

$$\equiv (p \wedge \neg q \wedge \neg r) \vee (p \wedge (q \vee \neg r)), \quad \text{by absorption law}$$

$$\equiv [(p \wedge \neg q \wedge \neg r) \vee p] \wedge [(p \wedge \neg q \wedge \neg r) \vee (q \vee \neg r)]$$

$$\equiv p \wedge [(p \wedge \neg q \wedge \neg r) \vee q], \quad \text{by absorption law}$$

$$\equiv p \wedge (q \vee \neg r), \quad \text{by absorption law}$$

10. Principal Normal Form

What Are Principal Normal Forms?

Principal normal forms are canonical ways to represent any propositional logic formula using only AND, OR, and NOT.

They come in two dual flavors:

1. **Principal Disjunctive Normal Form (PDNF):**

   A sum-of-products built from all true assignments.

2. **Principal Conjunctive Normal Form (PCNF):**

   A product-of-sums built from all false assignments.

10.1. Minterms and Principal Disjunctive Normal Form (PDNF)

Intuitive Idea

"Lock & Key"

Imagine a safe with a combination lock that only opens when you hold exactly the right set of keys.

Here, each key is like a variable (say p or q), and having or not having a key corresponds to True or False.

"Minterm"

A minterm is a very specific "recipe" of keys — all keys listed, either you have them (True) or you do not (False) — that makes the lock open.

"Principal Disjunctive Normal Form (PDNF)"

When we OR together every recipe that works, we get the Principal Disjunctive Normal Form (PDNF).

10.2. Maxterms and Principal Conjunctive Normal Form (PCNF)

Intuitive Idea

"Lock & Key"

Imagine a safe with a combination lock that only opens when you hold exactly the right set of keys.

Here, each key is like a variable (say p or q), and having or not having a key corresponds to True or False.

"Maxterm"

A maxterm is a very specific "barrier" of keys — that blocks the lock from opening.

Each barrier lists all variables, joined by OR, describing a situation where the formula is false.

"Principal Conjunctive Normal Form (PCNF)"

When we AND together every barrier (maxterm), we get the Principal Conjunctive Normal Form (PCNF).

10.3. Minterms vs Maxterms

Aspect	**Minterm**	**Maxterm**
Built from	True rows	False rows
Clause form	AND of literals	OR of literals
Combined via	OR	AND
Purpose	Lists satisfying cases	Blocks falsifying cases
Used in	PDNF	PCNF

Example 10.1. *Without constructing the truth tables, find the principal disjunctive normal forms of the following statements:*

1. $$(\neg p \to q) \wedge (q \leftrightarrow p)$$

2. $$(p \wedge q) \vee (\neg p \wedge q) \vee (q \wedge r)$$

3. $$p \wedge \neg(q \wedge r) \vee (p \to q)$$

4. $$(q \vee (p \wedge r)) \wedge \neg((p \vee r) \wedge q)$$

Solution (1)

$$(\neg p \to q) \wedge (q \leftrightarrow p)$$

$$\equiv (p \vee q) \wedge ((p \wedge q) \vee (\neg p \wedge \neg q))$$

$$\equiv ((p \vee q) \wedge (p \wedge q)) \vee ((p \vee q) \wedge \neg(p \vee q))$$

$$\equiv ((p \vee q) \wedge (p \wedge q)) \vee F$$

$$\equiv (p \wedge (p \wedge q)) \vee (q \wedge (p \wedge q))$$

$$\equiv (p \wedge q) \vee (p \wedge q)$$

$$\equiv p \wedge q$$

Solution (2)

$$(p \wedge q) \vee (\neg p \wedge q) \vee (q \wedge r)$$

$$\equiv ((p \wedge q) \wedge (r \vee \neg r)) \vee ((\neg p \wedge q) \wedge (r \vee \neg r)) \vee ((q \wedge r) \wedge (p \vee \neg p))$$

$$\equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r)$$

$$\vee (p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r)$$

$$\equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r)$$

(Deleting repetition of identical minterms)

Solution (3)

$$p \wedge \neg(q \wedge r) \vee (p \to q)$$

$$\equiv (p \wedge (\neg q \vee \neg r)) \vee (\neg p \vee q)$$

$$\equiv (p \wedge \neg q) \vee (p \wedge \neg r) \vee \neg p \vee q$$

$$\equiv (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \wedge (q \vee \neg q)) \vee (q \wedge (p \vee \neg p))$$

$$\equiv (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q) \vee (p \wedge q) \vee (\neg p \wedge q)$$

$$\equiv ((p \wedge \neg q) \wedge (r \vee \neg r)) \vee ((p \wedge \neg r) \wedge (q \vee \neg q))$$

$$\vee((\neg p \wedge q) \wedge (r \vee \neg r)) \vee ((p \wedge q) \wedge (r \vee \neg r))$$

$$\equiv (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (p \wedge q \wedge \neg r)$$

$$\vee(\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r)$$

$$\vee(p \wedge q \wedge \neg r)$$

$$\equiv (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (p \wedge q \wedge \neg r)$$

$$\vee(\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r)$$

(Omitting repetitions)

Note 10.2. *Since all possible minterms are present in the PDNF, we infer that the given statement is a tautology.*

Solution (4)

$$(q \vee (p \wedge r)) \wedge \neg((p \vee r) \wedge q)$$

$$\equiv (q \vee (p \wedge r)) \wedge (\neg(p \vee r) \vee \neg q)$$

$$\equiv (q \vee (p \wedge r)) \wedge ((\neg p \wedge \neg r) \vee \neg q)$$

$$\equiv (q \wedge \neg p \wedge \neg r) \vee (q \wedge \neg q) \vee (p \wedge r \wedge \neg p \wedge \neg r) \vee (p \wedge r \wedge \neg q)$$

$$\equiv (\neg p \wedge q \wedge \neg r) \vee F \vee F \vee (p \wedge \neg q \wedge r)$$

$$\equiv (\neg p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r)$$

(Deleting F's)

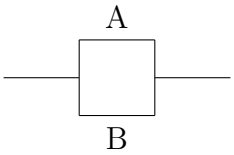
11. Applications of Boolean Algebra to Switching Circuits

11.1. Switching Circuits

Boolean algebra is widely used in digital electronics and switching theory.

- AND operation corresponds to series connection.
- OR operation corresponds to parallel connection.
- NOT operation corresponds to complement.

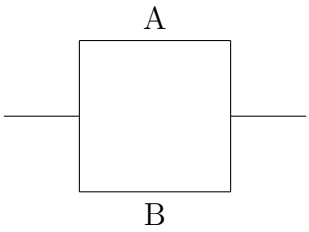
11.2. Series Circuit



The Boolean expression is:

$$AB$$

11.3. Parallel Circuit



The Boolean expression is:

$$A + B$$

12. Solved Examples

Example 12.1. *Simplify:*

$$A + AB$$

Solution:

$$A + AB = A(1 + B) = A$$

Example 12.2. *Find the complement of:*

$$A + B$$

Solution: *Using De Morgan's law,*

$$(A + B)' = A'B'$$

Example 12.3. *Convert into DNF:*

$$p \to q$$

Solution:

$$p \to q \equiv \neg p \vee q$$

Hence it is already in DNF.

Example 12.4. *Find the complement of each element of $(D_{42}, /)$. Also, prove that, in any distributive lattice $(S, \leq)$, the joint cancellation law holds, i.e., if $a \vee b = a \vee c$ and $a \wedge b = a \wedge c$ then $b = c$.* *(VIT-2025-2026 MTE)*

Solution: Complements in $(D_{42}, /)$ and Cancellation Law

For complements of elements in D_{42}:

The set of divisors of 42 is $D_{42} = \{1, 2, 3, 6, 7, 14, 21, 42\}$. In the lattice $(D_{42}, /)$, the meet $(\wedge)$ is defined as $\gcd(a, b)$ and the join $(\vee)$ is defined as $\text{lcm}(a, b)$. An element b is a complement of a if $a \wedge b = 1$ and $a \vee b = 42$.

Element (a)	*Complement (a')*	*Reasoning*
1	*42*	$\gcd(1, 42) = 1, \text{lcm}(1, 42) = 42$
2	*21*	$\gcd(2, 21) = 1, \text{lcm}(2, 21) = 42$
3	*14*	$\gcd(3, 14) = 1, \text{lcm}(3, 14) = 42$
6	*7*	$\gcd(6, 7) = 1, \text{lcm}(6, 7) = 42$
7	*6*	$\gcd(7, 6) = 1, \text{lcm}(7, 6) = 42$
14	*3*	$\gcd(14, 3) = 1, \text{lcm}(14, 3) = 42$
21	*2*	$\gcd(21, 2) = 1, \text{lcm}(21, 2) = 42$
42	*1*	$\gcd(42, 1) = 1, \text{lcm}(42, 1) = 42$

For Joint Cancellation Law
We have given $a \vee b = a \vee c$ *and* $a \wedge b = a \wedge c$ *in a distributive lattice* $(S, \leq)$.
To prove that $b = c$.

$$\begin{aligned}
b &= b \vee (b \wedge a) && \text{(By Absorption Law)} \\
&= b \vee (a \wedge b) \\
&= b \vee (a \wedge c) && \text{(Given } a \wedge b = a \wedge c\text{)} \\
&= (b \vee a) \wedge (b \vee c) && \text{(By Distributive Law)} \\
&= (a \vee c) \wedge (b \vee c) && \text{(Given } a \vee b = a \vee c\text{)} \\
&= (a \wedge b) \vee c && \text{(By Distributive Law)} \\
&= (a \wedge c) \vee c && \text{(Given } a \wedge b = a \wedge c\text{)} \\
&= c && \text{(By Absorption Law)}
\end{aligned}$$

Thus, $b = c$.

Example 12.5. *Express the following function into its Sum of Product (SOP) / pdnf without using truth table.* *(VIT-2024-2025 TEE)*

(a) $((xy)'z)'((x' + z)(y' + z'))'$

(b) $x + yz + xyz$

(c) $xz' + y'$

Solution: Sum of Products (SOP) / PDNF

$$\begin{aligned}
(a)\quad \text{Let } F &= ((xy)'z)'((x' + z)(y' + z'))' \\
&= [(xy)'' + z'] \cdot [(x' + z)' + (y' + z')'] && \text{(De Morgan's Law)} \\
&= (xy + z') \cdot (xz' + yz) \\
&= xyxz' + xyyz + z'xz' + z'yz \\
&= xyz' + xyz + xz' + 0 && \text{(Since } z'z = 0\text{)} \\
&= xyz' + xyz + xz'(y + y') \\
&= xyz' + xyz + xyz' + xy'z' \\
&= xyz + xyz' + xy'z' && \text{(By Idempotent Law)}
\end{aligned}$$

PDNF: $m_7 + m_6 + m_4 = \sum(4, 6, 7)$

(b) *Let* $F = x + yz + xyz$

$$= x(y + y')(z + z') + (x + x')yz + xyz$$
$$= (xyz + xyz' + xy'z + xy'z') + (xyz + x'yz) + xyz$$
$$= xyz + xyz' + xy'z + xy'z' + x'yz$$

PDNF: $m_7 + m_6 + m_5 + m_4 + m_3 = \sum(3, 4, 5, 6, 7)$

(c) *Let* $F = xz' + y'$

$$= xz'(y + y') + y'(x + x')(z + z')$$
$$= (xyz' + xy'z') + (xy'z + xy'z' + x'y'z + x'y'z')$$
$$= xyz' + xy'z' + xy'z + x'y'z + x'y'z'$$

PDNF: $m_6 + m_4 + m_5 + m_1 + m_0 = \sum(0, 1, 4, 5, 6)$

Example 12.6. *Construct the tables for* $\wedge$ *and* $\vee$ *for the set of all divisors of 36 under the relation of divisibility. Also show that it forms a lattice.* *(VIT-2024-2025 TEE)*

Solution: Let $D_{36} = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$. Here, $a \wedge b = \gcd(a, b)$ and $a \vee b = \text{lcm}(a, b)$. Meet Table ($\wedge$ or gcd)

$\wedge$	**1**	**2**	**3**	**4**	**6**	**9**	**12**	**18**	**36**
1	1	1	1	1	1	1	1	1	1
2	1	2	1	2	2	1	2	2	2
3	1	1	3	1	3	3	3	3	3
4	1	2	1	4	2	1	4	2	4
6	1	2	3	2	6	3	6	6	6
9	1	1	3	1	3	9	3	9	9
12	1	2	3	4	6	3	12	6	12
18	1	2	3	2	6	9	6	18	18
36	1	2	3	4	6	9	12	18	36

Join Table ($\vee$ or lcm)

$\vee$	**1**	**2**	**3**	**4**	**6**	**9**	**12**	**18**	**36**
1	1	2	3	4	6	9	12	18	36
2	2	2	6	4	6	18	12	18	36
3	3	6	3	12	6	9	12	18	36
4	4	4	12	4	12	36	12	36	36
6	6	6	6	12	6	18	12	18	36
9	9	18	9	36	18	9	36	18	36
12	12	12	12	12	12	36	12	36	36
18	18	18	18	36	18	18	36	18	36
36	36	36	36	36	36	36	36	36	36

It forms a Lattice, since a set forms a lattice if every pair of elements has a unique greatest lower bound (meet) and a unique least upper bound (join). Since for any $a, b \in D_{36}$, the operations $\gcd(a, b)$ and $\text{lcm}(a, b)$ yield distinct elements that are also members of D_{36}, it satisfies the definition and forms a lattice.

Example 12.7. *Construct an input/output table for the following Boolean function: $f(A, B, C) = (A + B)\bar{C}$, also find the disjunctive normal form of the Boolean function. Hence to represent the logical expressions with min term. (VIT-2024-2025 TEE)*

Solution: Step 1:

A	**B**	**C**	$\bar{\mathbf{C}}$	$\mathbf{A + B}$	$\mathbf{f(A, B, C) = (A + B)\bar{C}}$
0	0	0	1	0	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	0	1	0
1	1	0	1	1	1
1	1	1	0	1	0

Step 2: Disjunctive Normal Form (DNF): The Disjunctive Normal Form is the sum of the minterms for which the output of the function is 1. From the truth table, the function yields 1 for the following rows:

1. $A = 0, B = 1, C = 0 \implies \bar{A}B\bar{C}$
2. $A = 1, B = 0, C = 0 \implies A\bar{B}\bar{C}$
3. $A = 1, B = 1, C = 0 \implies AB\bar{C}$

Thus, the DNF is:

$$f(A, B, C) = \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C}$$

Step 3: Representation with Min terms Mapping the binary inputs to their decimal equivalents ($010_2 = 2$, $100_2 = 4$, and $110_2 = 6$), the expression can be written in min term notation as:

$$f(A, B, C) = \sum m(2, 4, 6)$$

In a bounded lattice, an element x is a complement of an element y if:

$$x \wedge y = \mathbf{0} \quad \text{(the least element)} \quad \text{and} \quad x \vee y = \mathbf{1} \quad \text{(the greatest element)}$$

From the provided Hasse diagram:

- The least element ($\mathbf{0}$) is e.
- The greatest element ($\mathbf{1}$) is f.

For element a: We look for an element x such that $a \wedge x = e$ and $a \vee x = f$.

- $a \wedge b = e$
- $a \vee b = f$

Thus, the complement of a is b.

For element c: We look for an element x such that $c \wedge x = e$ and $c \vee x = f$.

- $c \wedge e = e$, but $c \vee e = c \neq f$.
- $c \wedge b = e$, but $c \vee b = f$ holds. However, this is not a distributive lattice and b is already a complement of a. Looking closely at the connections, there is no single element that satisfies both the greatest lower bound and least upper bound properties strictly as a unique complement for c.

Thus, the element c does not have a complement in this lattice.

Solution 9 (b): The given switching network consists of two main branches connected in parallel between nodes B and L. We can represent parallel connections with addition ($+$) and series connections with multiplication ($\cdot$).

- Top Branch: Two sub-circuits are connected in series.
  - The first sub-circuit has switches p and q in parallel: $(p + q)$
  - The second sub-circuit has switches p and t in parallel: $(p + t)$
  - Combined expression for top branch: $(p + q) \cdot (p + t)$
- Bottom Branch: Switch t is in series with a sub-circuit containing switches p, q, and t all in parallel.
  - The sub-circuit expression: $(p + q + t)$

– Combined expression for bottom branch: $t \cdot (p + q + t)$

Since the top and bottom branches are in parallel, we add their expressions. Thus, the Boolean function representing the circuit is:

$$F(p, q, t) = [(p + q) \cdot (p + t)] + [t \cdot (p + q + t)] \tag{1}$$

Simplifying the Boolean Function
We can simplify the function using the standard laws of Boolean algebra:

$$F = (p + q)(p + t) + t(p + q + t)$$

Apply the distributive law to the first term $(p + y)(p + z) = p + yz$, and distribute t in the second term:

$$= (p + qt) + (tp + tq + t \cdot t)$$

Since $t \cdot t = t$:

$$= p + qt + pt + qt + t$$

Using the idempotent law, $qt + qt = qt$:

$$= p + pt + qt + t$$

Factor out t from the last three terms:

$$= p + t(p + q + 1)$$

Since $1 + \text{anything} = 1$ in Boolean algebra, the entire term in the parentheses becomes 1:

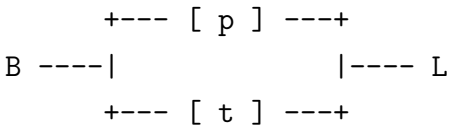
$$= p + t(1)$$
$$= p + t$$

The equivalent simplified Boolean function is:

$$F = p + t \tag{2}$$

Drawing the Simplified Circuit
The simplified function $F = p + t$ indicates that the original complex network is logically

equivalent to a simple circuit with only two switches, p and t, connected in **parallel**.



Example 12.8. *Find the simpler network corresponding to the Boolean expression* $F(a,b,c) = a.b.c + a.b^{'}.c + a^{'}.b^{'}.c^{'}$ *(VIT-2022-2023 TEE)*

Solution: The following steps are used

1. Observe that the first two terms ($a \cdot b \cdot c$ and $a \cdot b' \cdot c$) have common factors a and c.

$$F = (a \cdot c)(b + b') + a' \cdot b' \cdot c'$$

2. Apply the Complement Law:

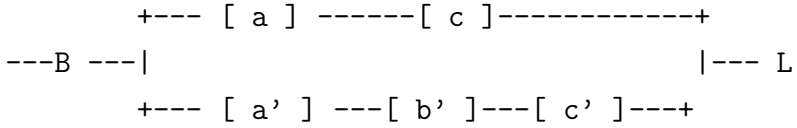
$$F = (a \cdot c)(1) + a' \cdot b' \cdot c'$$

3. Apply the Identity Law:

$$F = a \cdot c + a' \cdot b' \cdot c'$$

The simplified Boolean expression is:

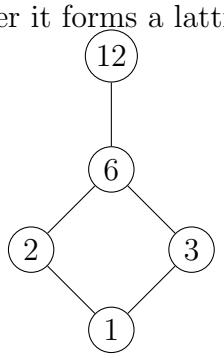
$$F(a,b,c) = ac + a'b'c' \tag{3}$$



Exercises: Lattices and Boolean Algebra

Lattice as POSET

1. Let $S = \{1, 2, 3, 6, 12\}$ be a set ordered by the divisibility relation ($a \leq b$ if a divides b). Draw the Hasse diagram for the partially ordered set $(S, \leq)$ and determine whether it forms a lattice.



Ans.:{ It is lattice.}

2. Let $X = \{a, b, c\}$ and consider its power set $\mathcal{P}(X)$ ordered by subset inclusion ($\subseteq$). Find the infimum (meet) and supremum (join) for the elements $\{a\}$ and $\{b, c\}$.
Ans.: {

$$\begin{aligned}\text{infimum}(\{a\}, \{b, c\}) &= \{a\} \cap \{b, c\} \\ &= \emptyset\end{aligned}$$

$$\begin{aligned}\text{supremum}(\{a\}, \{b, c\}) &= \{a\} \cup \{b, c\} \\ &= \{a, b, c\} = X\end{aligned}$$

}

Distributive Lattice

3. Prove that every chain (a totally ordered set) is a distributive lattice.
Ans.: { Hint: $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ holds true in all possible order configurations, the distributive identity is satisfied. Therefore, every chain is a distributive lattice.}

4. Show that the diamond lattice M_3 (the five-element lattice containing a top element, a bottom element, and three mutually incomparable elements between them) is not distributive.
Ans.: {Hint: Since LHS $= a$ and RHS $= 0$, we have:

$$a \wedge (b \vee c) \neq (a \wedge b) \vee (a \wedge c)$$

Because the distributive law fails for these elements, the diamond lattice M_3 **is not**

distributive.}

Boolean Algebra as Lattice

5. Prove that in any Boolean algebra, the complement of every element is unique.
   Ans.: {Hint: Since $y = z$, the initial assumption that x can have two distinct complements is false. Therefore, the complement of every element in a Boolean algebra is unique.}

6. Let $D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$ be the lattice of divisors of 30 under the divisibility relation. Verify whether D_{30} constitutes a Boolean algebra.
   Ans.: {Hint: Because D_{30} is a complemented distributive lattice with 2^3 elements, it **does constitute a Boolean algebra**. (Specifically, it is isomorphic to the power set Boolean algebra $\mathcal{P}(\{2, 3, 5\})$).}

Laws of Boolean Algebra

7. Using the algebraic laws of Boolean algebra (such as De Morgan's laws and absorption laws), simplify the following expression:

$$A \cdot (\bar{A} + B) + B \cdot (B + C)$$

   Ans.: {The simplified form of the Boolean expression is: $[B]$}

8. Prove the Consensus Theorem algebraically:

$$X \cdot Y + \bar{X} \cdot Z + Y \cdot Z = X \cdot Y + \bar{X} \cdot Z$$

   Ans.: {Since the Left-Hand Side simplifies directly to the Right-Hand Side, the Consensus Theorem is algebraically prove.}

Boolean Functions & Normal Forms

9. Find the Principal Disjunctive Normal Form (PDNF), also known as the Sum-of-Minterms, for the Boolean function:

$$f(X, Y, Z) = (X + Y) \cdot (\bar{X} + Z)$$

   Ans.: {The Principal Disjunctive Normal Form (PDNF) is:

$$\bar{X}Y\bar{Z} + \bar{X}YZ + X\bar{Y}Z + XYZ$$

}

Application to Switching Circuits

10. Design a switching circuit for a hallway light controlled by two separate switches (one at the entrance and one at the exit) such that flipping either switch toggles the state of the light. State its Boolean expression.
Ans.{

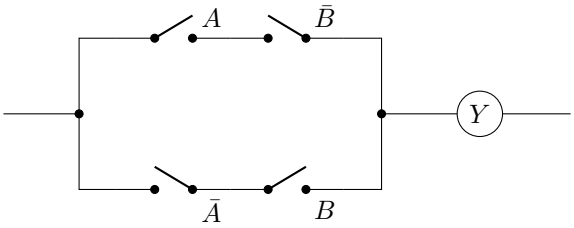
1. Functional Logic (Truth Table): Let A represent the status of the switch at the entrance, and B represent the switch at the exit. Let Y represent the state of the hallway light (1 for ON, 0 for OFF).

The requirement states that flipping either switch must toggle the light's current state. This operation corresponds exactly to the **Exclusive-OR (XOR)** logic gate function:

A	B	Y	Description
0	0	0	Initial state (Light is OFF)
0	1	1	Switch B is flipped (Light turns ON)
1	0	1	Switch A is flipped (Light turns ON)
1	1	0	Both switches are flipped (Light turns OFF)

2. Boolean Expression: Using the Sum-of-Products (SOP) form derived from the rows where $Y = 1$:

$$Y = A \oplus B = A\bar{B} + \bar{A}B$$



The network configuration satisfies the toggling condition: current flows through to the lamp Y if and only if exactly one of the two switches is in its alternative state.

}